

Last name _____

First name _____

LARSON—MATH 310—HOMEWORK WORKSHEET 07
Matrices!

1. Write up a **neat** assignment on a **new sheet** of paper. (Do not cram your answers between the lines).
2. **Number** your problems so that it is easy to see what work matches the assigned problems.
3. Be verbose. Remember that you do not understand a concept if you do not know an **examples**.

Problems

1. Let the columns of matrix A be $\vec{a}_1, \dots, \vec{a}_6$.

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- (a) Find $\vec{a}_1, \dots, \vec{a}_6$.
 - (b) Why can you instantly say that the columns are linearly dependent?
 - (c) What is the largest possible number of linearly independent columns? (Cite the appropriate result from class. Explain.).
2. Let M be the 3×2 matrix:
$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$
 - (a) Write out $M_{i,j}$ for $i = 1, 2, 3$ and $j = 1, 2$.
 - (b) Write M as a matrix of column vectors $\begin{bmatrix} \vec{a}_1 & \vec{a}_2 \end{bmatrix}$. What are $\vec{a}_1$ and $\vec{a}_2$?
 - (c) Write M as a matrix of row vectors vectors $\begin{bmatrix} \vec{b}_1 \\ \vec{b}_2 \\ \vec{b}_3 \end{bmatrix}$. What are $\vec{b}_1$, $\vec{b}_2$ and $\vec{b}_3$?
3. What is the matrix $0_{3,2}$? Is it the same as $0_{2,3}$?
4. Find the diagonal matrix $\text{diag}(3, 4, 5)$.
5. Find a 3×3 *upper triangular* matrix U with all non-zero entries equal to 2.
6. Find a 3×3 *lower triangular* matrix L with all non-zero entries equal to 3.

7. Let M be the 3×2 matrix:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

- (a) Find the *transpose* M^T .
- (b) Find the scalar product $5M$.
- (c) Let N be the 3×2 matrix:

$$\begin{bmatrix} -1 & 0 \\ 3 & 4 \\ 7 & 8 \end{bmatrix}$$

Find $M + N$.

8. (a) What is the *definition* of $A\vec{x}$, for an $m \times n$ matrix A and a $n \times 1$ vector $\vec{x}$?

Let A be the 3×2 matrix: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$. Let $\vec{x} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$.

- (b) Use this definition of matrix-vector multiplication to find $A\vec{x}$.
- (c) Use the *row interpretation* of matrix-vector multiplication to find $A\vec{x}$.
- (d) Use the *column interpretation* of matrix-vector multiplication to find $A\vec{x}$.
- (e) Find $A\vec{e}_1$.
- (f) Find $A\vec{e}_2$.

9. A be the 3×2 matrix: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$

We'll see that $f(\vec{x}) = A\vec{x}$ defines a *linear* function $f: \mathbb{R}^2 \rightarrow \mathbb{R}^3$.

We first check that f is additive.

- (a) Let $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Find $A\vec{x}$.
- (b) Let $\vec{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$. Find $A\vec{y}$.
- (c) Find $\vec{x} + \vec{y}$.
- (d) Find $A(\vec{x} + \vec{y})$.
- (e) Find $A\vec{x} + A\vec{y}$. Check that $A\vec{x} + A\vec{y} = A(\vec{x} + \vec{y})$.

Now we check that f is homogenous.

- (f) Let α be a scalar. Find $\alpha\vec{x}$.
- (g) Find $A(\alpha\vec{x})$.
- (h) Find $\alpha(A\vec{x})$. Check that $A(\alpha\vec{x}) = \alpha(A\vec{x})$.